

电力电子

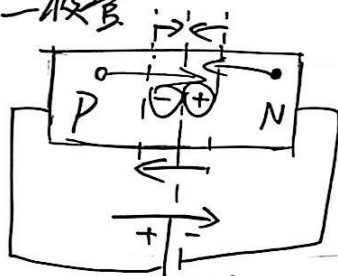
(第4版)

技术基础

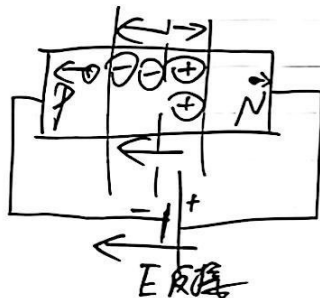
张凯锋 吴晓梅 包金明 / 编著
冷增祥 徐以荣

电力电子 Chapter 1 晶闸管

二极管

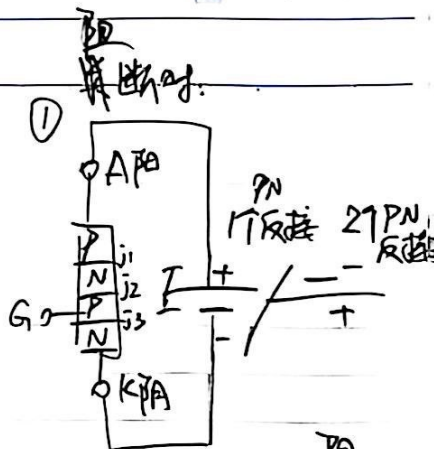


E 正接



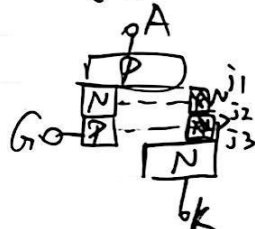
E 反接

晶闸管
SCR
模型



三个半导, 总有反接 阻 断时.

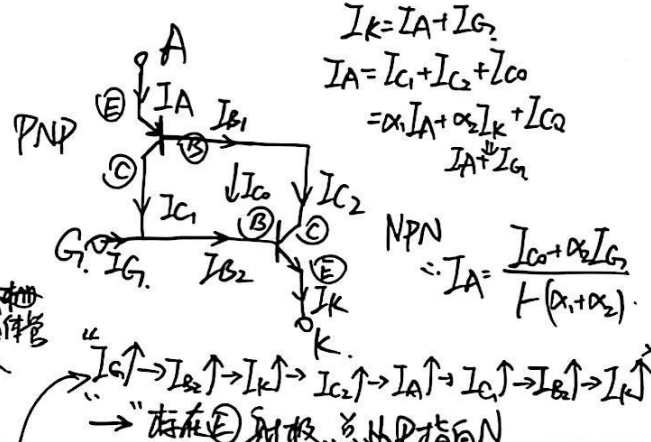
② 导通时:



PNP & NPN
2个三极管复合

功率三极管

IGBT
MOSFET
GTR
PRT
晶闸管
SCR
GTO



$$I_K = I_A + I_G$$

$$I_A = I_{C1} + I_{C2} + I_{C0}$$

$$= \alpha_1 I_A + \alpha_2 I_K + I_{C0}$$

$$I_A + I_G$$

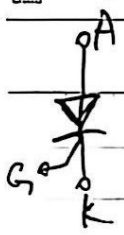
$$I_A = \frac{I_{C0} + \alpha_2 I_K}{1 - (\alpha_1 + \alpha_2)}$$

“ $I_{C1} \rightarrow I_{B2} \rightarrow I_{C2} \rightarrow I_{A1} \rightarrow I_{C1}$ ”
“存在①射极, 总从P指向N”

场效应管
MOSFET
“半导”

MOSFET 中同

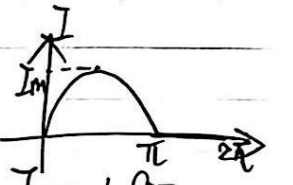
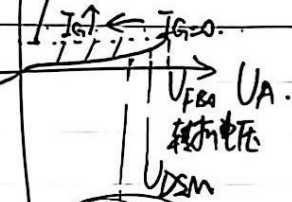
伏安特性:



I_A



$$I_{VT} = \frac{I_m}{2} = \frac{\pi I_T}{2}$$



$$I_T = \frac{1}{2\pi} \int_0^{2\pi} I_m \sin \omega t d\omega t$$

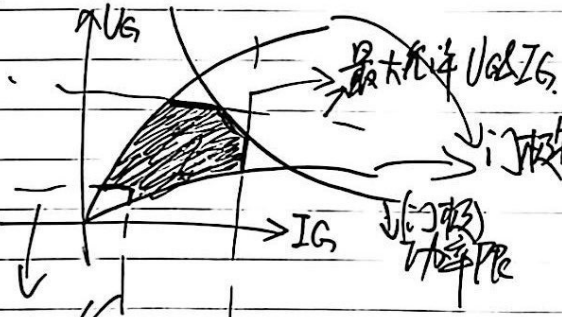
$$= \frac{I_m}{\pi} \int_0^{\pi} \sin \omega t d\omega t = \frac{I_m}{\pi}$$

DLR/S 可重复峰值
RRR/R 可重复峰值

min 为
 U_T 额定电压

* 选型 I_T : 取峰值 I_m 乘 2/3 倍
实际波形

门极特性:



$$I_{VT} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_m^2 \sin^2 \omega t d\omega t}$$

$$= I_m \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\omega t}{2} d\omega t$$

$$= I_m \frac{1}{2} \frac{\pi}{2} = \frac{I_m}{2}$$

$$= \frac{\pi I_T}{2}$$

(SCR) 波形系数: 用于选型 I_T

$$K = \frac{I_{VT}}{I_{d(s)}}$$

U_{GT} 温度下
能触发的
 $\min U_G$ & I_G

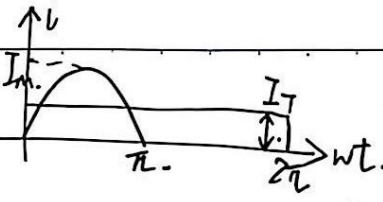
* I_T 额定电流: 环境温度 +40°C 且在规定的冷却条件下, 电阻负载, 正弦半波, 导通角 $\geq 170^\circ$, 结温稳定且额定结温所允许的最大平均电流

2 (变压器) 波形系数 $K = I_{VT}/I_d$, 用于选铜线直径.

I_{VT} 相等 I_T .

选型: SCR U_T & I_T 要
2~3 倍

eg.1 实际:



又对波形系数也选
(仅 I_T 定义为 I_{VT} 不同)
而选在功率定下不用再 $\times 2$
表在功率定 S 及例 1.3 节

实际: $I_m = 100A$

额定:

$$I_{VT} = 50A = 100 \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \omega t d\omega t}$$

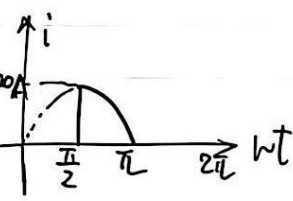
$$\Leftrightarrow I_{VT} = 50A$$

$$I_{VT} = \frac{I_m}{2} = \frac{\pi I_T}{2}$$

$$I_d = \frac{100}{\pi} A = \frac{100}{\pi} \int_0^{\pi} \sin \omega t d\omega t = \frac{100}{\pi}$$

$$I_T = \frac{100}{\pi} A$$

eg.2: 实际:



实际: $I_m = 100A$

额定:

$$I_{VT} = 100 \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \omega t d\omega t}$$

$$= \frac{100}{2} = 50A$$

eg.4: $I_T = 100A$ 则 $I_{VT} = 50\pi A$

$$\downarrow \therefore 50\pi = I_m \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \omega t d\omega t} = \frac{I_m}{2}$$

$$\therefore I_m = 100\pi$$

$$\therefore I_d = \frac{100\pi}{\pi} = 100A$$

$$I_{VT} = 25\sqrt{2}A = \frac{I_m}{2} = \frac{\pi I_T}{2}$$

$$\therefore I_T = \frac{50\sqrt{2}}{\pi} A$$

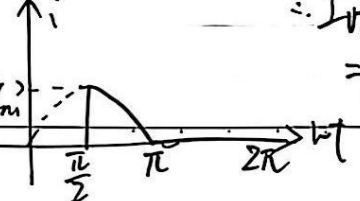
(反解)

eg.3: $I_T = 100A$ 求 I_d 允许? $I_T = 100A$

$$\therefore I_{VT} = 50\pi A$$

$$\therefore I_d = \frac{100\pi}{\pi} = 100A$$

实际:



$$= I_m \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 \omega t d\omega t} = 70.7A$$

$$= \frac{I_m}{2} \therefore I_m = 100\sqrt{2}A$$

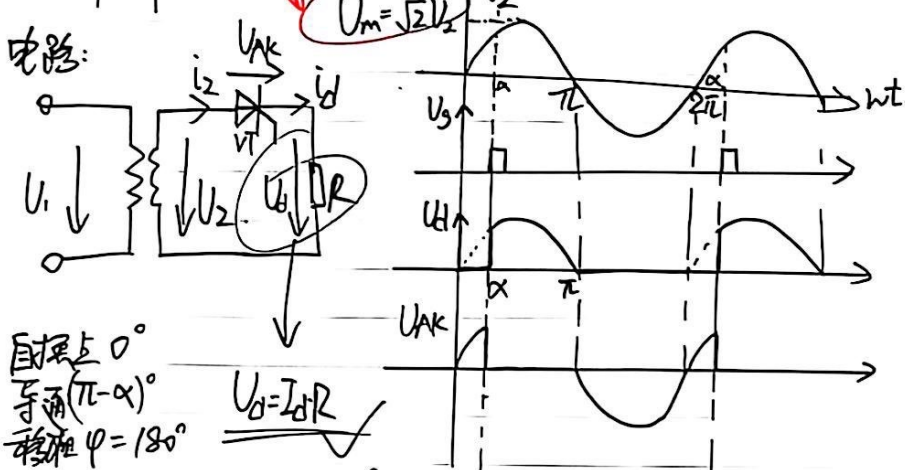
KOKUYO

整流 AC-DC 单相 - 半波 - 带续流

Tip: U_2 为变压器副边有效值
 $P_{0 \sim 12}$ 表 1.3
 单相半波 电阻负载

1° 单相半波

电路:



自然点 0°
 导通 $(\pi - \alpha)^\circ$
 移相 $\varphi = 180^\circ$

$$U_d = I_d R$$

$$U_d = \frac{1}{2\pi} \int_{\alpha}^{\pi} \sqrt{2} U_2 \sin \omega t d\omega t = \frac{U_2}{\sqrt{2}\pi} (\cos \alpha + 1)$$

$$I_{VT} = \frac{U_{VT}}{R} = \frac{\sqrt{2} U_2}{R} \int_{\alpha}^{\pi} \frac{1}{2\pi} \sin \omega t d\omega t = \frac{\sqrt{2} U_2}{R} \left[\frac{\pi - \alpha}{2\pi} + \frac{\sin 2\alpha}{4\pi} \right]$$

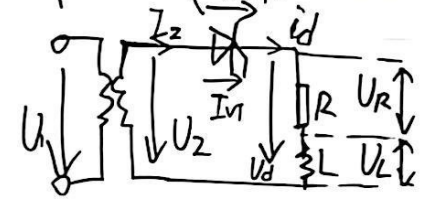
$$= \frac{U_2}{R} \left[\frac{\pi - \alpha}{2\pi} + \frac{\sin 2\alpha}{4\pi} \right]$$

$$K = \frac{I_{VT}}{I_d} = \frac{\frac{\sqrt{2} U_2}{R} \left[\frac{\pi - \alpha}{2\pi} + \frac{\sin 2\alpha}{4\pi} \right]}{\frac{U_2}{\sqrt{2}\pi} (\cos \alpha + 1)} = \frac{\sqrt{2} \left[\frac{\pi - \alpha}{2} + \frac{\sin 2\alpha}{2} \right]}{1 + \cos \alpha}$$

$$(\alpha = 0, K = \frac{\pi}{2} = 1.57)$$

$$I_2 = I_{VT}$$

电阻负载



$$U_2 = U_{VT} + U_L$$

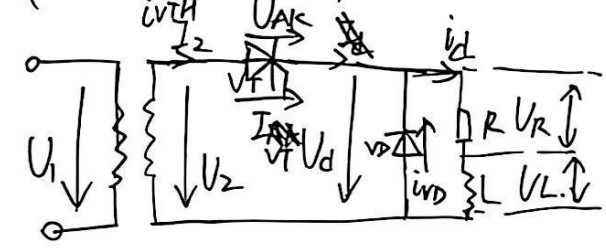
$$= U_2 R + L \frac{di_L}{dt}$$

i_2 与 U_L 波形相似

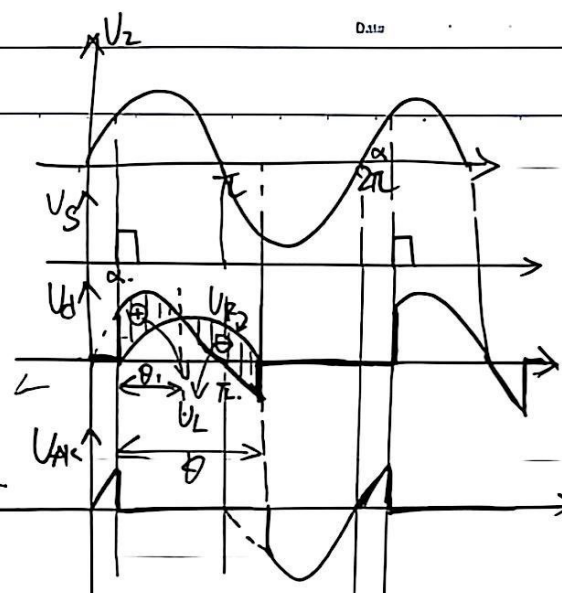
电感若太大, $\theta \uparrow$
 $\theta \rightarrow \pi - \alpha$
 则 $\theta_m \rightarrow 2(\pi - \alpha)$
 则 U_d 平均 $\rightarrow 0$

无输出 电压
 电源与电感
 能量互相转换

为解决: 加续流二极管
 (电感大, 认为电流平直)

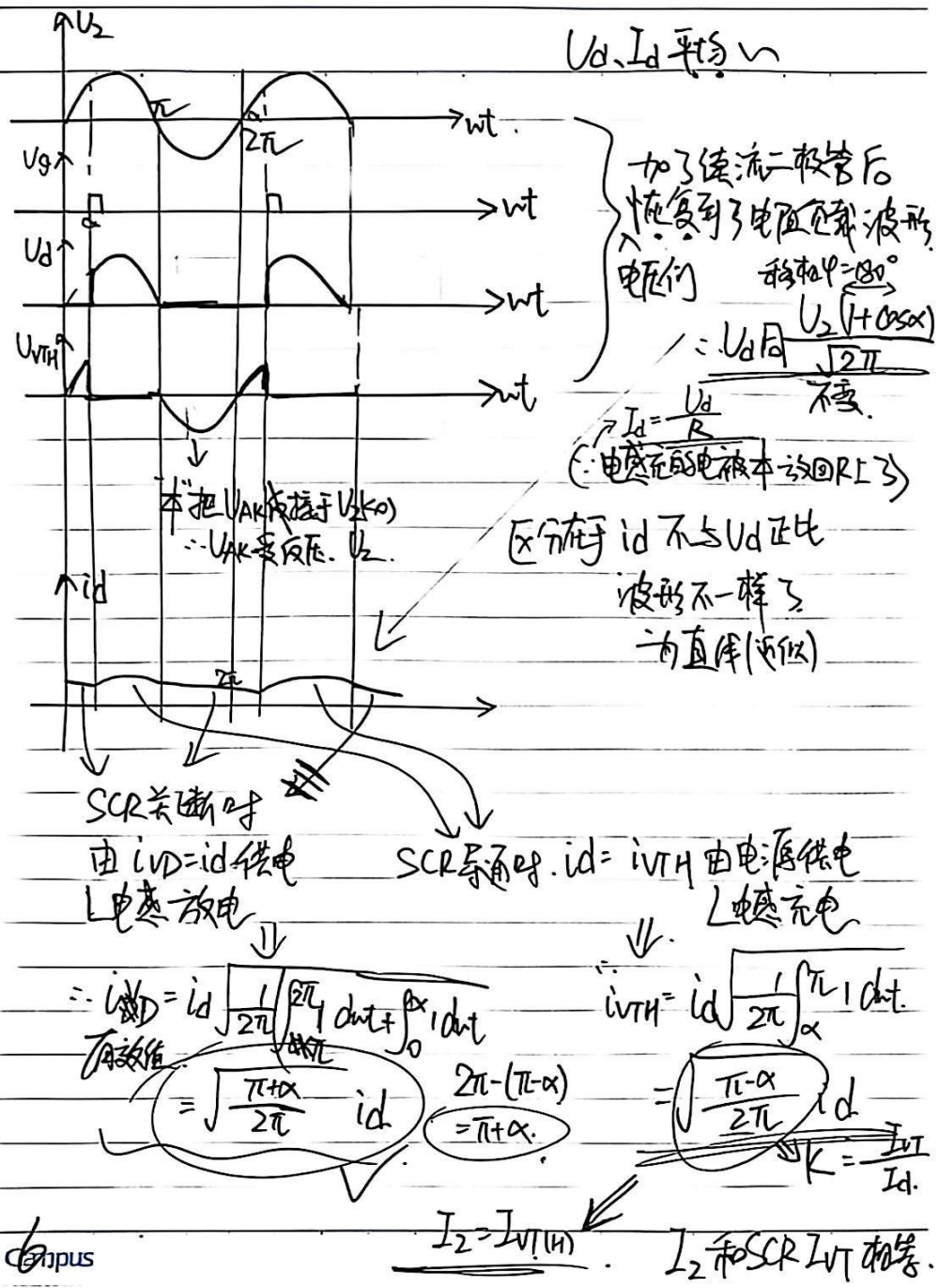


$U_2 \gg \Delta U_D \Rightarrow$
 $U_2 \ll \Delta U_D \Rightarrow$

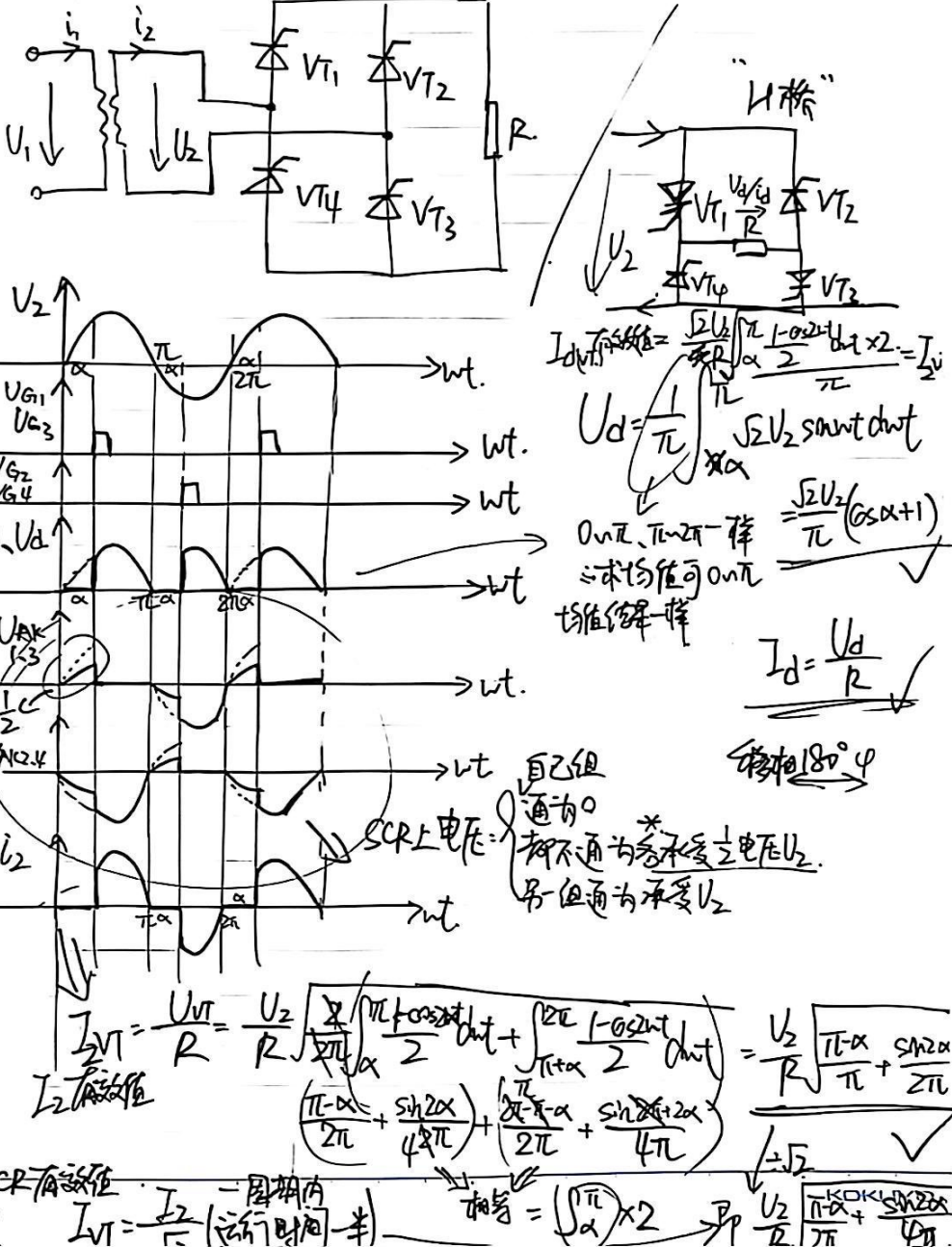


$$U_d = \frac{1}{2\pi} \int_{\alpha}^{\pi + \theta} \sqrt{2} U_2 \sin \omega t d\omega t$$

$$= \frac{\sqrt{2} U_2}{2\pi} (\cos \alpha - \cos(\alpha + \theta))$$



单相半桥 { 优点: 电路简单, SCR管少.
缺点: 输出电压小, i_2 方向不变 $\Rightarrow$ 滤波困难 直流磁化, 效率低



变压器副边SCR电流

I_2, I_{VT} 均指有效值

单相全控桥与单相半波不同: $I_2 \neq I_{VT} \Rightarrow I_2 = \sqrt{2} I_{VT}$

波形系数 k_f = 变压器二次侧 续流电流

$$k_f = \frac{\frac{U_2}{R} \sqrt{\frac{\pi-\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}}{\frac{\sqrt{2} U_2}{\pi} (1+\cos \alpha)} = \frac{\sqrt{\frac{\pi(\pi-\alpha)}{2} + \frac{\pi \sin 2\alpha}{2}}}{\sqrt{2} (1+\cos \alpha)}$$

晶闸管: $k_f = \frac{I_2}{\sqrt{2} I_d} = \frac{k_f}{\sqrt{2}} = \frac{\sqrt{\frac{\pi(\pi-\alpha)}{2} + \frac{\pi \sin 2\alpha}{2}}}{\sqrt{2} (1+\cos \alpha)}$

$I_{dVT} = I_{2VT} = I_{VT} \times \sqrt{2}$

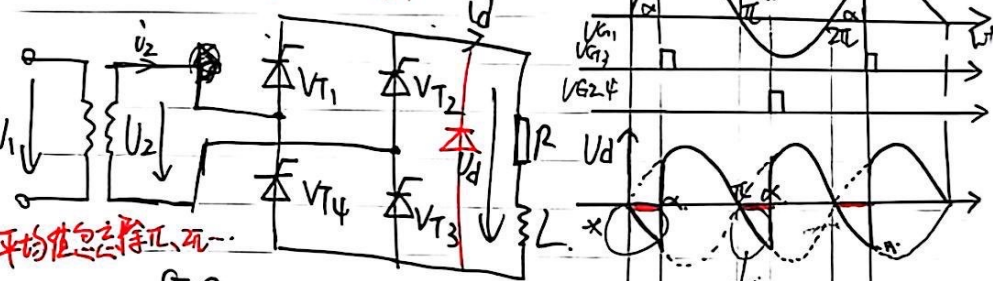
有效值 = 有效值 = $\sqrt{2}$ 倍晶闸管有效值

有效值 $I_{VT} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i_d^2 dt}$

有效值 $I_{VT} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i_d^2 dt}$

阻感负载: i_d 波形恢复原样

若加续流管: 电压SCR续流管被 i_d 替代



平均值 = $\frac{1}{\pi} \int_0^{\pi} i_d dt$

$$U_d = \sqrt{2} U_2 \int_0^{\pi} \sin \omega t d\omega t = \frac{\sqrt{2} U_2}{\pi} \int_0^{\pi} \sin \omega t d\omega t$$

$$= \frac{\sqrt{2} U_2}{\pi} (\cos \alpha + \cos(\alpha + \pi)) = \frac{\sqrt{2} U_2}{\pi} \cos \alpha$$

若不计算: $\frac{\sqrt{2} U_2}{\pi} \cos \alpha$

若计算: $\frac{\sqrt{2} U_2}{\pi} \cos \alpha$

若计算: $\frac{\sqrt{2} U_2}{\pi} \cos \alpha$

无论电路有无负载 P_{6002} 表 1.34 $I_d = \frac{U_d}{R}$ 恒成立

求平均电流 I_d 先求平均电压 U_d 除 R

移相范围 $0 \sim 90^\circ$ φ 如何理解?

$U_d = \frac{\sqrt{2} U_2}{\pi} \cos \alpha$

$\alpha = 90^\circ$ 已有 max & min; $0 \sim 180^\circ$

而其他电路/电阻负载时 $\varphi = 180^\circ$ 是因为 U_d 有 $1 + \cos \alpha$

$\varphi = 90^\circ$ $\alpha = 90^\circ$ 时 $U_d = 0$

由波形易得: 有效值 $U_d = i_2 = \sqrt{2} I_{VT}$

电流 负载 电阻 电感

$k = \frac{I_{VT}}{I_d} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ 恒值

eg. 1.3 单相全控桥式. 大 L 负载 $R = 2\Omega$. 输入交流电压 $60V$

1° 求 U_d 0.9 60

$U_d = \frac{\sqrt{2} U_2}{\pi} \cos \alpha = 54 (\cos \alpha) = 54 \text{ V}$

2° 选型SCR: $U_T = (2 \sim 3) \times U_m = (2 \sim 3) \cdot \sqrt{2} \cdot 60 = 169.7 \sim 254.5 \text{ V}$

$I_T = \frac{\sqrt{2} I_{VT}}{\pi} = \frac{\sqrt{2}}{\pi} \cdot \frac{i_d}{\sqrt{2}} = \frac{\sqrt{2}}{\pi} \cdot \frac{U_d}{R} = 12.15 \text{ A}$

选 200V 即可

3° 电源变压器容量 S :

$S = U_2 I_2 = 60 \times I_d = 60 \times \frac{U_d}{R} = 1020 \text{ (V} \cdot \text{A)}$

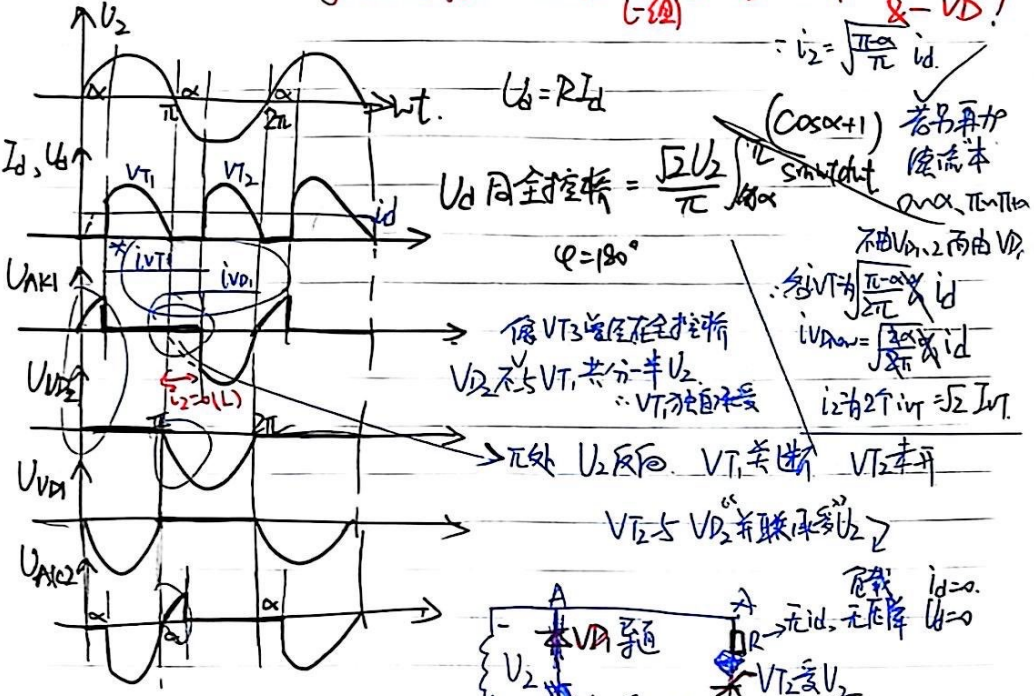
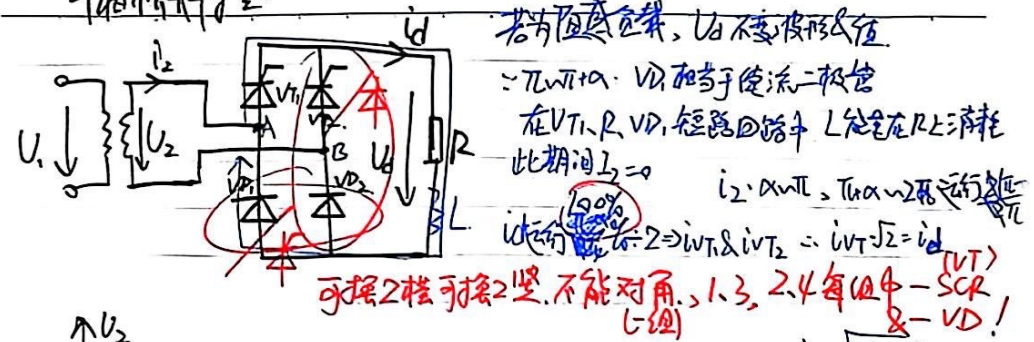
均为有效值

若电感 L 较小, 则续流管断续

$U_d = \frac{1}{\pi} \int_0^{\pi} \sqrt{2} U_2 \sin \omega t d\omega t = \frac{\sqrt{2} U_2}{\pi} (\cos \alpha - \cos(\alpha + \pi))$

★ P41 若 V_{T2} 信号错误, 则 V_{T1} 、 S_{VD1} 、 V_{D2} 循环损耗, “失控”现象, 要另加续流电感。
 半控 & 带续流二极管: 只能 $AC \rightarrow DC$ 单向电能传递

3° 单相桥式半控:



$I_2 = \sqrt{2} I_{VT}$
 一个周期内导 2 段 I_2
 每段分别只属于一个区间

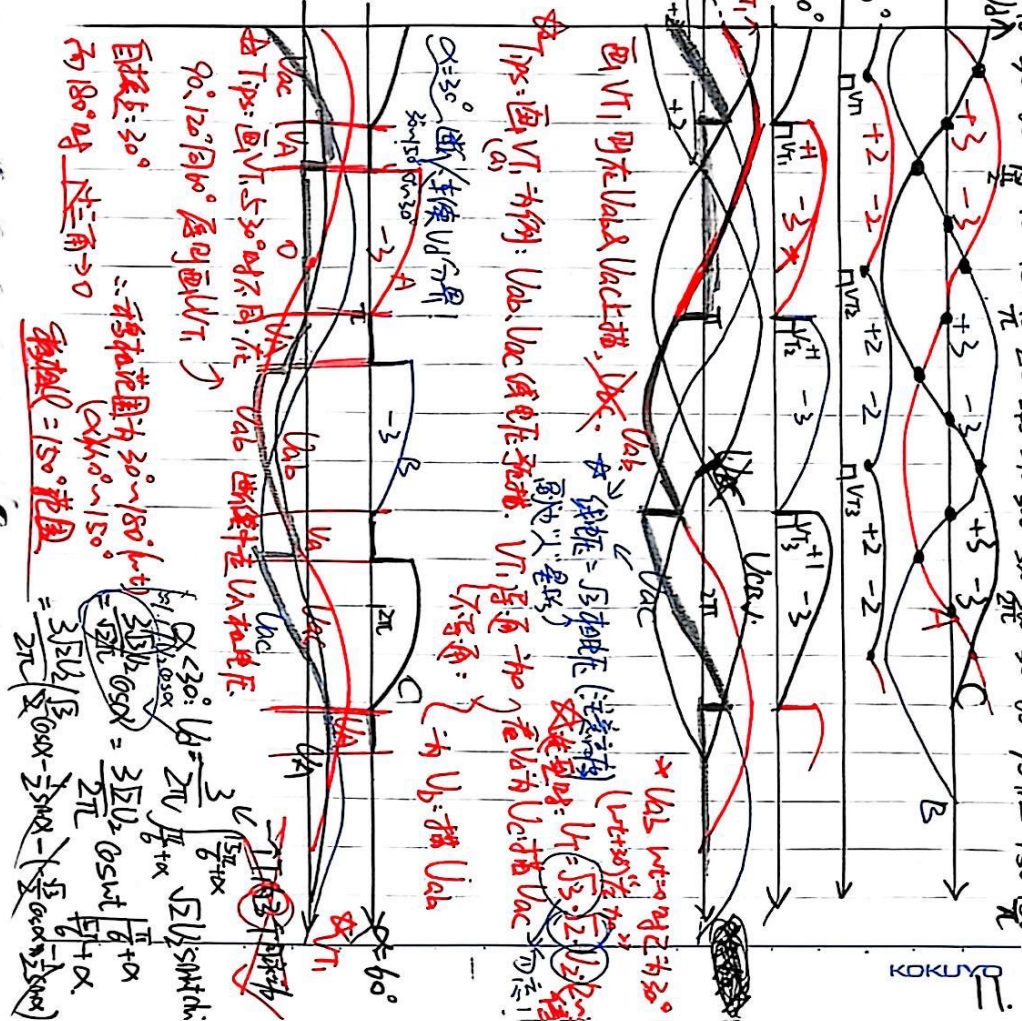
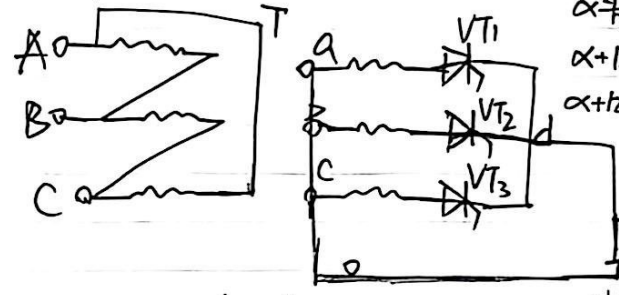
$$I_{VT} = \frac{U_2}{R} \frac{1}{\pi} \int_{\alpha}^{\pi} \sin \omega t d\omega t$$

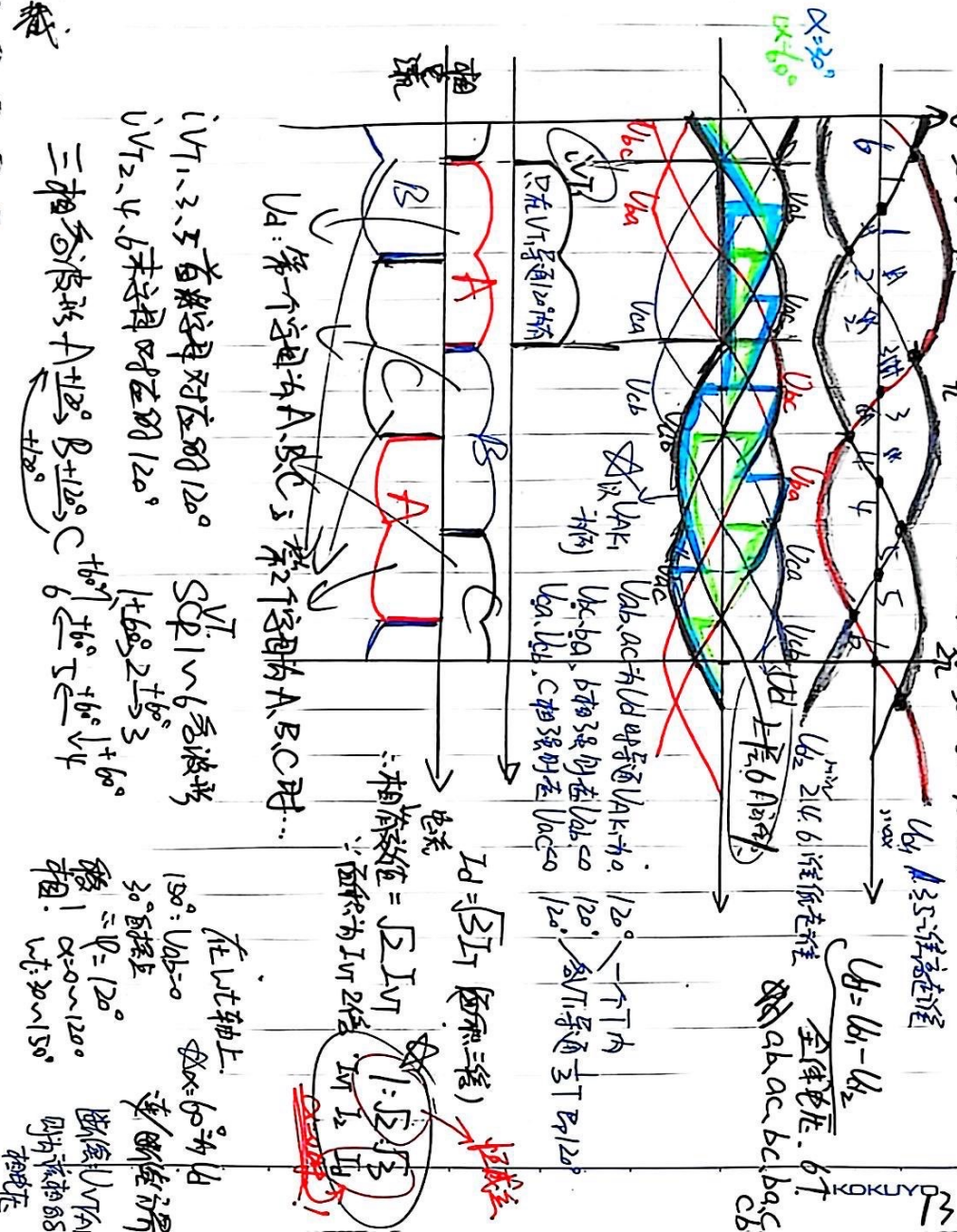
$$= \frac{U_2}{R} \left[\frac{\pi - \alpha}{\pi} + \frac{\sin 2\alpha}{2\pi} \right]$$

$$k = \frac{I_{VT}}{I_d} = \frac{\frac{U_2}{R} \left[\frac{\pi - \alpha}{\pi} + \frac{\sin 2\alpha}{2\pi} \right]}{\frac{\sqrt{2} U_2}{\pi R} (1 + \cos \alpha)}$$

$$= \frac{\pi(\pi - \alpha) + \sin 2\alpha}{2(1 + \cos \alpha)}$$

三相半波: 适用于大功率, 直流 U 脉动小, 易滤波。
 三相全桥应用最广。





$\alpha < 60^\circ$ $U_d = \frac{6}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sqrt{3} \cdot \sqrt{2} U_2 \sin(\omega t + 30^\circ) d\omega t = \frac{\sqrt{6} U_2}{\pi} \cos \omega t \Big|_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha}$

$U_d = \frac{\sqrt{6} U_2}{\pi} \cos \alpha$

易知: 统一为 ωt 坐标, 则 U_{ab} 应为 $\uparrow$

$= \frac{\sqrt{6} U_2}{2} \cos \omega t + 30^\circ \Big|_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} = \frac{\sqrt{6} U_2}{2} (\cos(60^\circ + \alpha) - \cos(90^\circ + \alpha))$

$= \frac{\sqrt{6} U_2}{2} \cos \alpha$

$\Rightarrow \frac{\sqrt{6}}{\pi} \approx 2.34$

$\alpha > 60^\circ$:

$U_d = \frac{6}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sqrt{3} \cdot \sqrt{2} U_2 \sin(\omega t + 30^\circ) d\omega t = \frac{\sqrt{6} U_2}{\pi} \cos \omega t + 30^\circ \Big|_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha}$

$= 2.34 U_2 (1 + \cos(60^\circ + \alpha))$

$(\cos(60^\circ + \alpha) = \cos 60^\circ \cos \alpha - \sin 60^\circ \sin \alpha)$

$\alpha < 60^\circ$ $I_{VT} = \frac{\sqrt{6} U_2}{R} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sin(\omega t + 30^\circ) d\omega t = \frac{\sqrt{6} U_2}{R} \left(-\cos(\omega t + 30^\circ) \Big|_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \right)$

$= \frac{\sqrt{6} U_2}{R} \left(\frac{1}{2} \frac{\pi}{3} + \frac{1}{4} \sin 2\omega t + 60^\circ \right)$

$= \frac{\sqrt{6} U_2}{R} \left(\frac{1}{2} \frac{\pi}{3} + \frac{1}{4} \sin 2\omega t + 60^\circ \right)$

$= \frac{\sqrt{6} U_2}{R} \left(\frac{1}{2} \frac{\pi}{3} + \frac{1}{4} \sin 2\omega t + 60^\circ \right)$

$60^\circ \sim 120^\circ$

$\alpha > 60^\circ$

$I_{VT} = \frac{\sqrt{6} U_2}{R} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sin(\omega t + 30^\circ) d\omega t = \frac{U_2}{R} \left(\frac{2\pi}{3} - \frac{3\alpha}{\pi} + \frac{3 \sin(2\alpha + 120^\circ)}{2\pi} \right)$

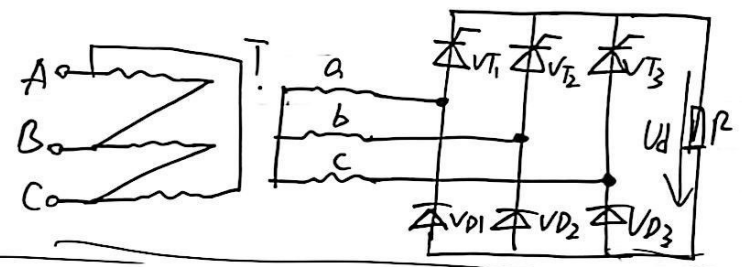
$\sin(2\alpha + 120^\circ) = \sin(2\alpha + 120^\circ)$

三相全桥

大电感负载下: U_2 与 电阻负载 $\alpha = 60^\circ$ 一致为 $2.34 U_2 \cos \alpha$

电流比例 $I_d : I_{VT} : I_2 = \sqrt{3} : 1 : \sqrt{2}$

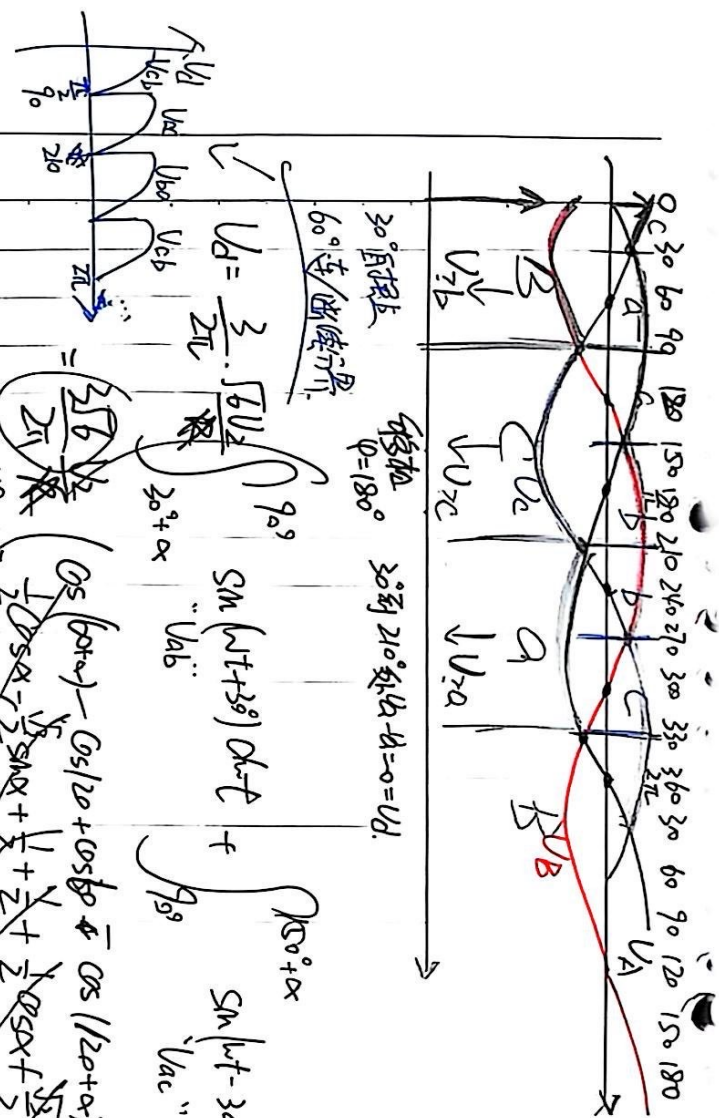
6° 三相半控桥



$\alpha < 60^\circ$

$I_{VT} = \frac{\sqrt{6} U_2}{R} \left(\frac{1}{2\pi} \int_{\frac{\pi}{6}+\alpha}^{\frac{\pi}{2}+\alpha} \sin^2(\omega t + 30^\circ) d\omega t + \int_{\frac{\pi}{2}+\alpha}^{\frac{5\pi}{6}+\alpha} \sin^2(\omega t - 30^\circ) d\omega t \right)$

$= \frac{\sqrt{6} U_2}{R} \left(\frac{1}{2\pi} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{4} (1 + \cos \alpha) \right) \right)$



$\alpha > 60^\circ$ 时

$$I_{VT} = \frac{\sqrt{6}U_2}{R} \frac{1}{2\pi} \int_{30+\alpha}^{210} \sin^2(\omega t - 30^\circ) d\omega t = \frac{\sqrt{6}U_2}{R} \sqrt{\frac{1}{2\pi} \left(\frac{\pi - \alpha}{2} + \frac{\sin 2\alpha}{4} \right)}$$

$= 1.17 \frac{U_2}{\sqrt{2}} (\cos \alpha + 1)$

$U_d = \frac{3}{2\pi} \int_{30+\alpha}^{90} \sqrt{6}U_2 \sin(\omega t + 30^\circ) d\omega t + \int_{150+\alpha}^{180} \sqrt{6}U_2 \sin(\omega t - 30^\circ) d\omega t$

$= \frac{3\sqrt{6}U_2}{2\pi} \left(\cos(60+\alpha) - \cos(120+\alpha) + \cos(60-\alpha) - \cos(120-\alpha) \right)$

$= 1.17 \frac{U_2}{\sqrt{2}} (\cos \alpha + 1)$

★ 对于 I_{VT} , I_d , I_2 以及有效值、平均值、面积的关系的更正!!!

1. 电阻负载下, 积分求 U_d , $I_d = \frac{U_d}{R}$ 恒有
无论电路/负载类型

半波电路 $I_2 = I_{VT}$
桥电路 $I_2 = \sqrt{2} I_{VT}$

在半波中流过晶闸管电流即流过交流二次侧绕组电流。
而在桥中, I_2 将正反各流一次 I_{VT}
故有效值 $\sqrt{2I^2} = \sqrt{2} I$

而 $\frac{I_{VT}}{I_d} \rightarrow$ 都是实积分
式中出现 α 是正常。

此外非面积比 = 有效值比!
 $\sqrt{2}$ 单纯由于 I_2 是 $\pm I_{VT}$ 相反数, 只差符号

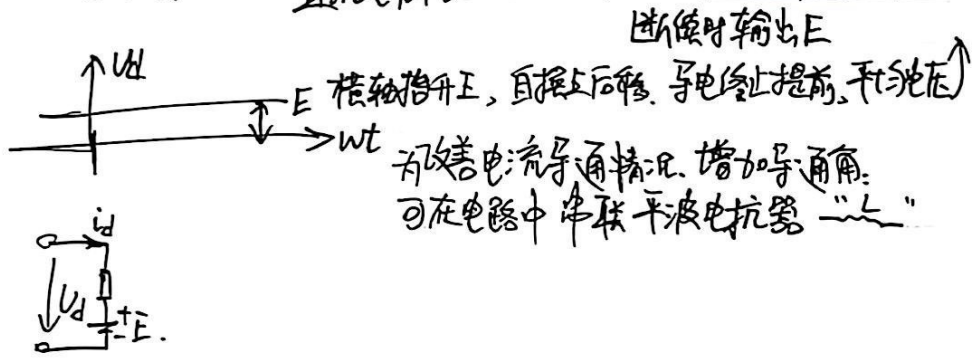
$k = \frac{I_{VT}}{I_d}$ 波形系数与 α 相关
(I_d 波形面积 = 3 $\cdot$ I_{VT} 波形面积)
 $\therefore I_d = \sqrt{3} I_{VT}$ X 错误

2. 大电感负载下: I_d, I_{VT}, I_2, I_{VD} 都近似直线, 水平波形
eg 桥电路连续波形, 可有经典 $I_{VT} = I_2 = I_d$ 比例
eg 单相中有 $\sqrt{\frac{\alpha}{\pi}} \times \sqrt{\frac{\pi - \alpha}{2\pi}}$... 等经典比例
续流管

可视为 I_d 的 $\frac{2}{2\pi}$ 是由 I_{VT}, I_{VD} 所构成。
最后比例 $\rightarrow$ 有效值 $\sqrt{2} \dots I_d / I_{VT}$
具体电路具体分析波形

★ 易错: U_2 为副边有效值 $\therefore$ 积分时相电压峰值 $\sqrt{2} U_2$!
“续”续电压再乘 $\sqrt{3} = \sqrt{6} U_2$!!!

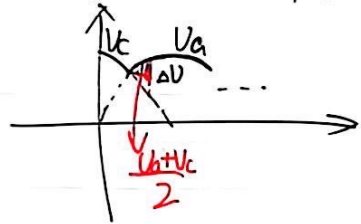
反电动势负载: 电池
直流电动机 } 有源负载



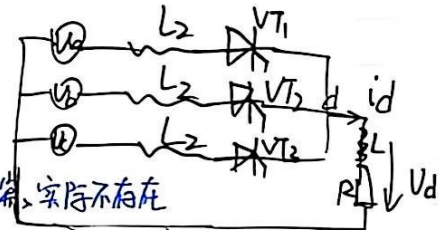
Chapter 2 变流器运行

2.1. 换向重叠角 γ (变压器漏电感, 铜导线 L, 进线电抗...)

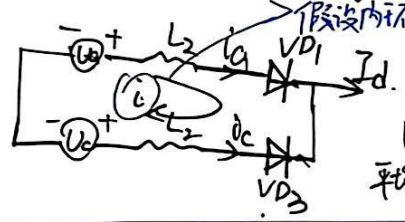
∴ 交流侧有电抗, 相续中电流不可突变, 有一段换流时间。



eg 换流时 $V_{T1}, 3$ 同时通
则 $i_d = i_a + i_c$



原理: $U_a - U_c = L_2 \frac{di_a}{dt} + L_2 \frac{di_c}{dt}$



eg 三相半波
 $U_d = \frac{3}{2\pi} \int_{\frac{\pi}{6}}^{\frac{\pi}{2} + \gamma} (U_a - \frac{U_a + U_c}{2}) d\omega t$

2.4 谐波

三相半波 3 个脉冲 (3k+1) 次谐波
桥 6 个 (6k+1)=2

- 危害
1. 谐波损耗, 发电/输电/用电效率 ↓
 2. 影响电气设备正常工作; 电机振动/噪音/过热; 变压器局部严重过热; 电抗 C、电抗过热; 绝缘老化; 寿命 ↓
 3. 电网中引发并/串联谐振, 放大谐波使 1, 2 加重
 4. 继电保护 & 自动装置误动作
 5. 干扰邻近通信 sys. 产生噪声/信息丢失

☆ n 次谐波含有率

$$HRI_n = \frac{I_n}{I_1} \times 100\%$$

☆ 电流总畸变率 $THD = \frac{\sqrt{\sum_{n=2}^N I_n^2}}{I_1} \times 100\%$ (有效值从 $n=2$ 到 N , 基波之外所有谐波)

2.5 功率因数

功率因数 $\lambda = \cos \varphi$ (功率因数)

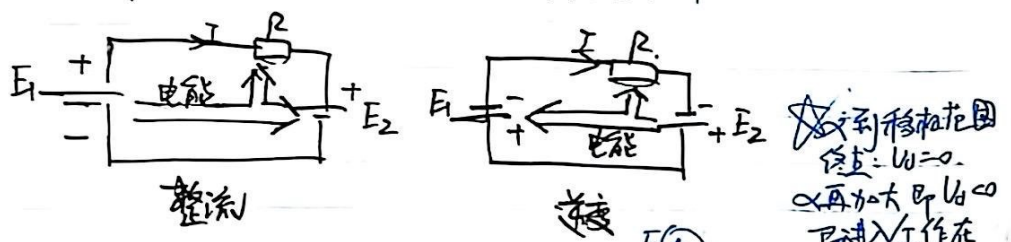
$S^2 = P^2 + Q^2$

$S = UI$ (kVA)
 $P = UI \cos \varphi$ (kW)
 $Q = UI \sin \varphi$ (S-P)

电压超前电流 $\lambda = \cos \varphi$
电压滞后电流 $\lambda = \cos \varphi$
电压超前电流 $\lambda = \cos \varphi$
电压滞后电流 $\lambda = \cos \varphi$

有源: 交流侧与电网相接
无源: 不与电网相接

① E_1, E_2 不可直接 ② 续流率不可变



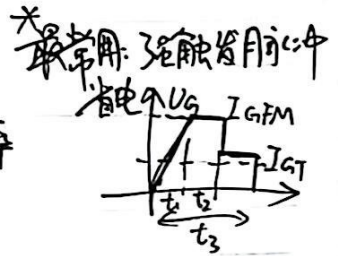
"有反大" 直流向交流侧逆变且 $|E| > |U_d|$

有一个直流电源 E
 E 的极性与整流电压(反)
逆变角/超前角 β $\alpha + \beta = 180^\circ$
 $\beta = 180^\circ, \alpha = 180^\circ$
与整流电压极性相反
e.g. U_d 均值为负数, α 较大
则工作在有源逆变
 $\beta >$ 整流电压角 $\gamma +$ 恢复角 δ
 $\beta = 25^\circ \sim 30^\circ$

Chapter 3 3.1 门极信号(触发)

3.5 定相

3.1 触发电路要求
脉冲应有一定幅值 & 功率
有一定宽度
前沿要陡
与主电路同步且有一定移相范围

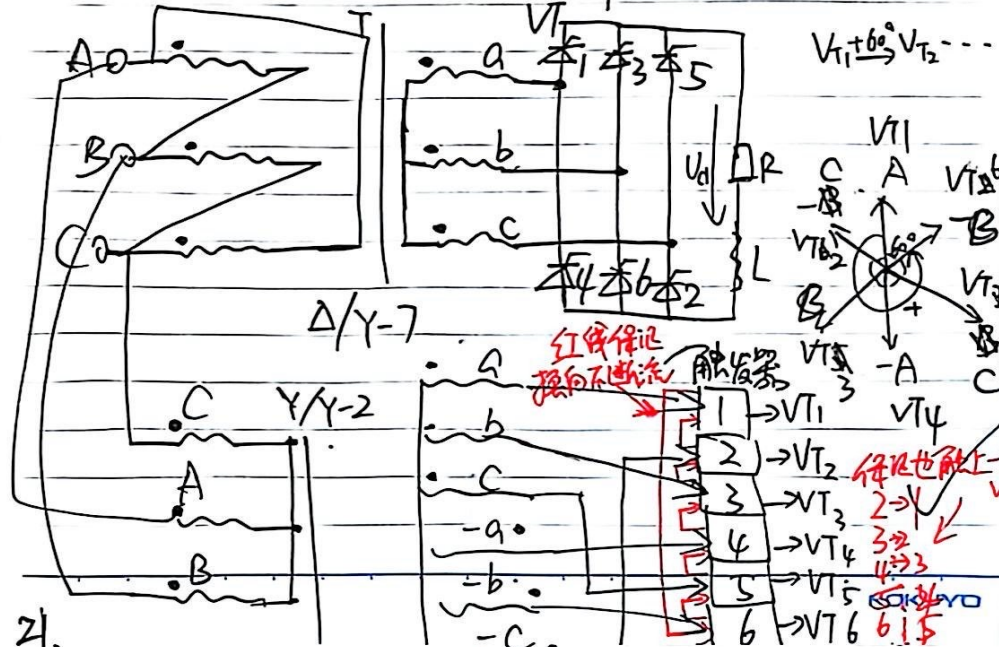
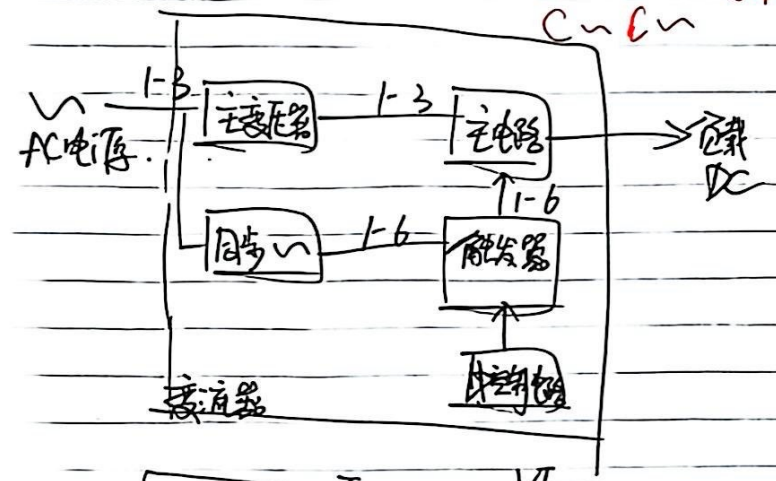


门极触发信号
DC 能耗大, 不用
AC 不稳定, 效率低
(U, I)
电压源
电流源

尖脉冲
宽脉冲
锯齿波

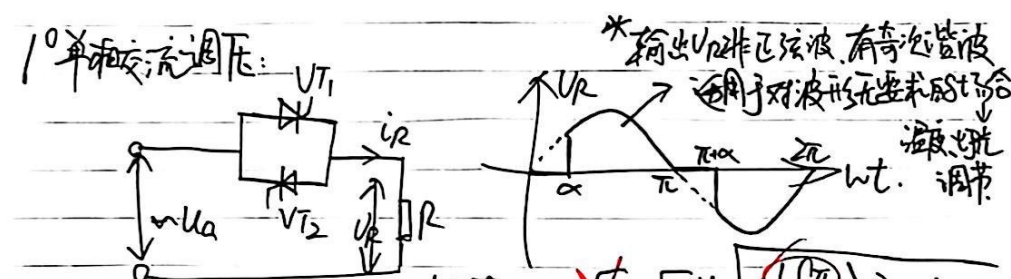
★ 触发电路
3.5 定相
时钟表示法 $\Delta/Y-n$
e.g. $30^\circ \sim 360^\circ$
表示原力基相比副力基超前 $n \times 30^\circ$

A 和 B 和 C 超前
B ~ B ~ 字母对应
C ~ C ~



Chapter 4 AC-AC

1. 调压 (相位控制)
 2. 调功 (通断控制)
 3. AC-AC 直接
 4. AC-DC-AC 间接



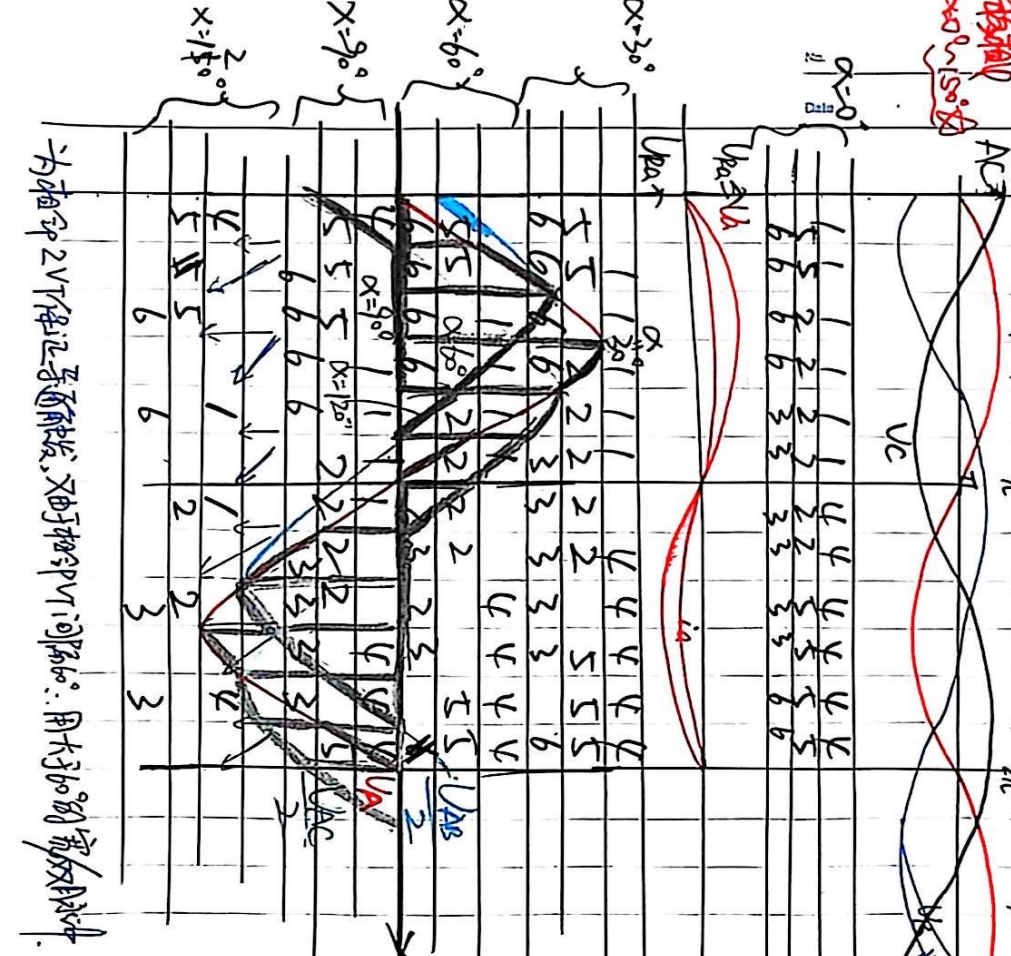
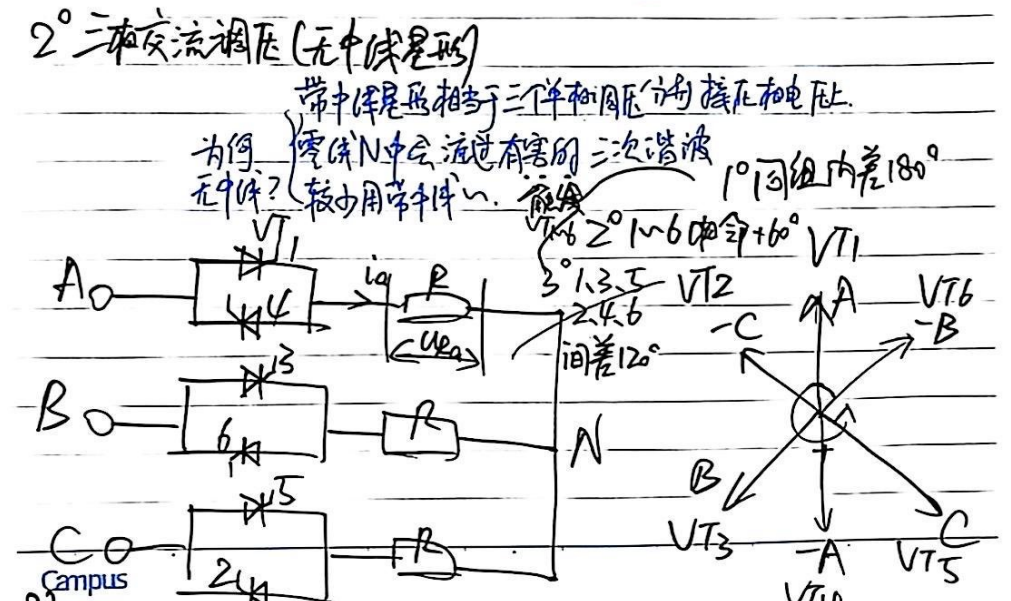
$$I_R = \frac{U_p}{R} = \frac{U_a}{R} \sqrt{\frac{2\pi - 2\alpha + \sin 2\alpha}{2\pi}}$$

$$U_p = \sqrt{2} U_a \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} \sin^2 \omega t d\omega t}$$

$$U_p = \sqrt{2} U_a \sqrt{\frac{\pi - \alpha}{2\pi} + \frac{\sin 2\alpha}{4\pi}}$$

输出有功功率: $P_R = U_p I_R = \frac{U_a^2}{R} \left(\frac{2\pi - \sin 2\alpha}{2\pi} \right)$

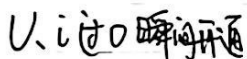
$$= U_a \sqrt{1 - \frac{2\alpha - \sin 2\alpha}{2\pi}}$$



1° 晶闸管导通顺序
 2° $\alpha = 0 \sim 60^\circ$, 电路处于 3 个晶闸管、2 个二极管同时导通的交替状态。
 每个晶闸管从 0° 到 30° (共 180°)
 2° $\alpha = [60^\circ \sim 90^\circ]$, 任何时刻为 2 晶闸管导通, 导通 120° 。
 3° $\alpha = [90^\circ \sim 150^\circ]$, 2 晶闸管、1 晶闸管、1 二极管交替导通, 导通 $300^\circ - 2\alpha$ 角, 但始终为 2 段, 导通角 $180^\circ - 150^\circ$ 。
 4° 晶闸管导通顺序: 1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6...
 5° 晶闸管导通顺序: 1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6...
 6° 晶闸管导通顺序: 1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6...

騎

主电路用交流调压, 但不用双相相位控制, 而用通断控制



波形为正弦波、无谐波/优点
干扰

$$V_o = \sqrt{\frac{1}{T_c} \int_0^{MT} V_n^2 dt} = \sqrt{\frac{MT}{T_c}} V_n$$

T_c 内所有周波都导通输出的电压有效值

改N

改功率及输出电压有效值

① T₂ 加权像: 正常血管中都可采通!

V-A特性为普通VT的曲

线条要居中对称至

AC
字面作为初始值

As a result, the AVR has a high degree of accuracy.

100 150 200 250 300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000

$$j27 \parallel 47 = \frac{j27 \times 47}{j27 + 47} = \frac{j1269}{47 + j27} \quad \text{反并联}$$

单极: 多子(载流子)
(空穴型)

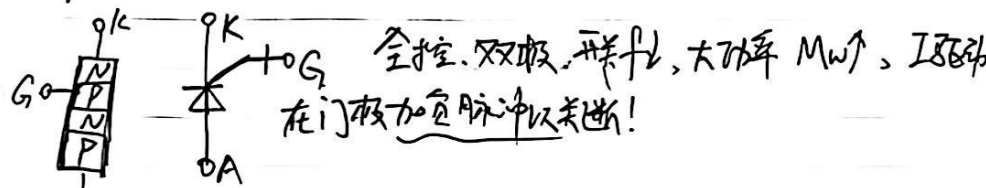
大功率晶体管 BJT/PRT/GTR

申优可夜应晶体管 MOSFET

绝缘栅双极晶体管

碳化硅 SiC、氮化镓 GaN

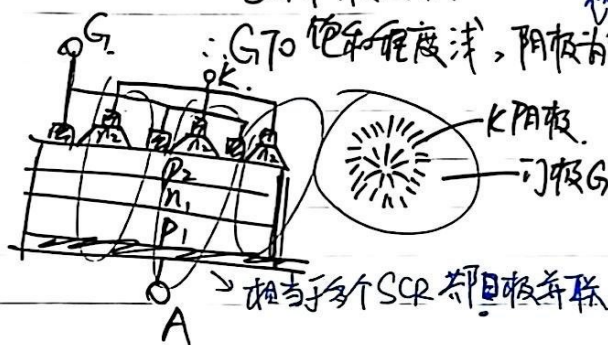
1° GTO



SCR 阴极S太大

被门极包围的

饱和程度浅, 阳极为首北小岛 $\rightarrow$ 越靠越易关闭
(G 阳极越大)

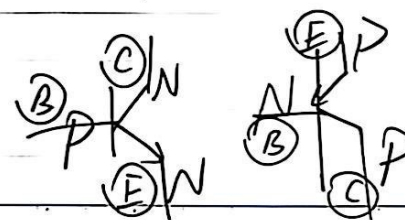
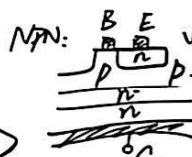


2° JGTR 大功率晶体管

二极管、双极、单极、场、耐压

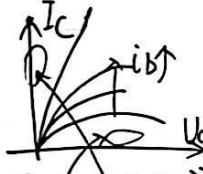
~

NPN:  中小功率代替SCR



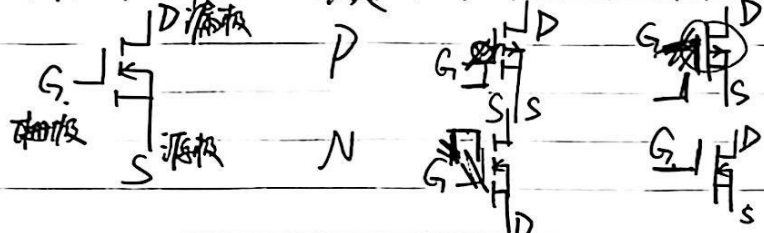
箭头在 (E) 上且从 P 指向 N

“大反三招”工作于禁止在8个区和

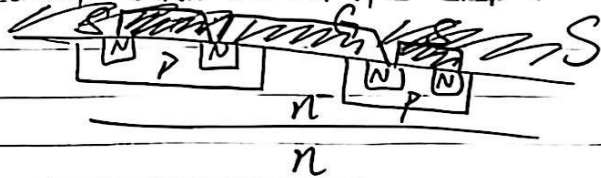


GT_D^S D=DC
S=CE

3° P-MOSFET



单极, $V_{GS} > 0$, 开关 $\uparrow$, I_A 耐压 $\downarrow$, 功率 $< 10\text{kw}$

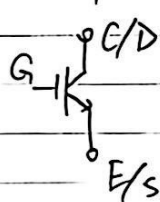


★增强型N沟道MOSFET

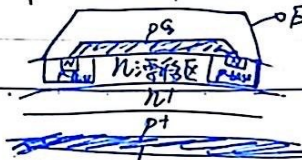


$V_{GS} > 0$ 形成耗尽层, $V_{GS} > V_{TH}$ 形成反型层进入导通, N沟道

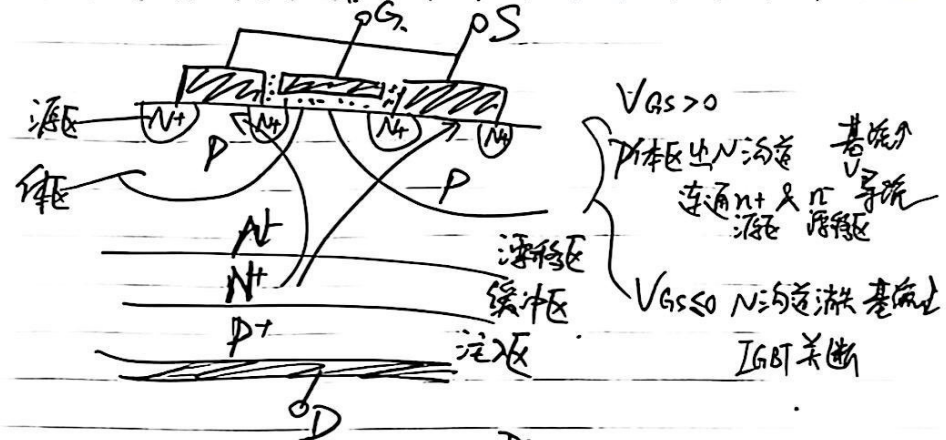
4° IGBT



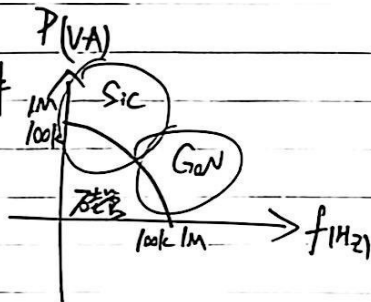
双极, $V_{GS} > 0$, 开关 $\uparrow$, I_A 耐压 $\uparrow$
“以MOSFET为驱动元件, GTR为主导元件的复合管”



IGBT由栅极控制开/断



5° SiC & GaN 更适合功率器件
 $\downarrow$ High Power $\downarrow$ High Speed



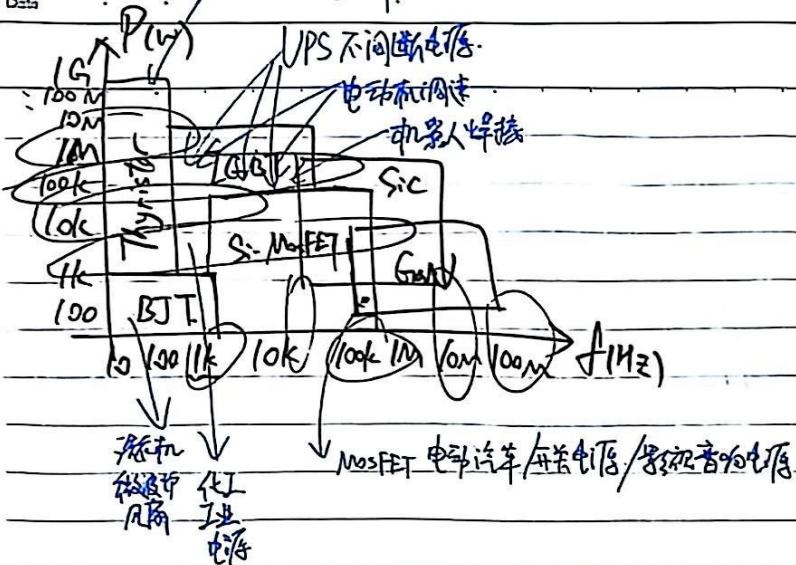
器件横向比较:

IGT: $\uparrow$ 控制频率
U: $\uparrow$ 控制频率, 控制功率 $\downarrow$

双极: 耐压 $\uparrow$, 耐压 $\downarrow$, 电流容量大, GTO, IGBT
单: 开关特性好, 驱动方便, MOSFET
复合: 双, 单极优点, IGBT

直流输电

IGBT.



二极管：整流、续流、能量回馈

G7R 被IGBT替代

SCR——整流

GTO 大體被

MosFET DC-DC 整流

IGBT DC-DC 變.

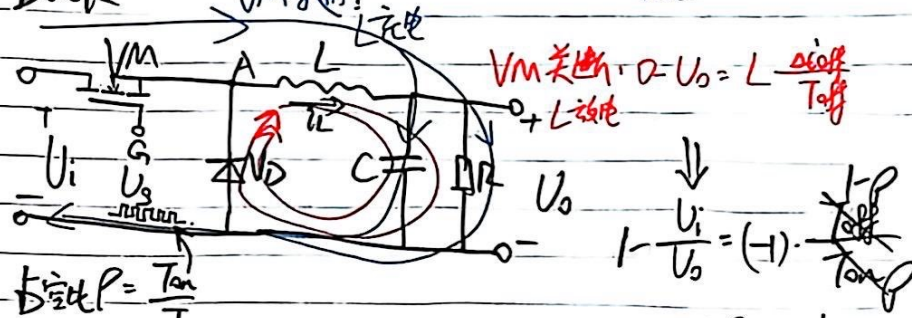
Data

Chapter 6. DC-DC

证明核心原则: 稳态时 $\Delta i_{on} + \Delta i_{off} = 0$ 电感L

1^o Buck

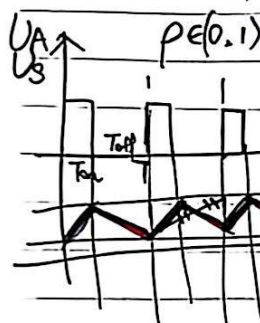
$$V_i = V_o = 2 \frac{\Delta i_{on}}{I_{on}}$$



$$1 - \frac{v_i}{v_o} = (-1) \cdot \frac{1-p}{1-p}$$

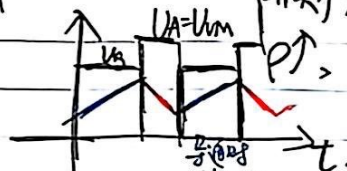
$$\tan \rho = \frac{T_{\text{an}}}{T}$$

$$\therefore \frac{V_i}{V_o} = 1 + \frac{1-p}{p} = \frac{1}{p}$$



$U_0 < U_i$
降压电路 $\therefore U_0 = pU_i \quad p \in (0, 1)$ ✓

Tips: 1. VM 封装器件
 电感太小时能量在T管中耗尽会断流
 开关1个可使电感减小，并不过耗尽又充
 电了



文字潦草
 p 越大 U_0 越大 但 $U_0 = p U_i$
 $0 < p < 1$
 降压电路 U_0 小于 U_i

2° Boost



Tips: 电感有电压泵升作用
储存电能短时间释放

$$\Rightarrow V_i = V_o(1 - \rho)$$

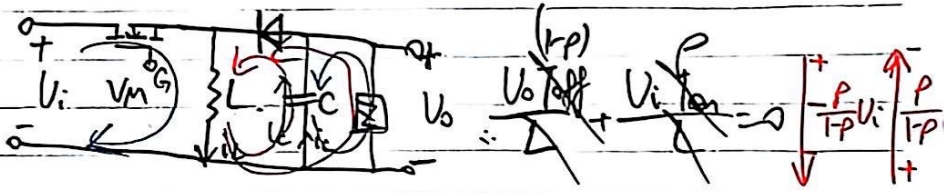
$$\therefore V_o = \frac{1}{1-\rho} V_i$$

VM 类 乐通

V_M 关断: $V_i - V_o = L \frac{di_{off}}{T_{off}}$
L 放电

$$\frac{U_i - U_0}{V} + \frac{(U_i - U_0) \times (1 - P)}{V} = 0$$

3° Buck-Boost: VM导通: $U_o = L \frac{\Delta i_{on}}{T_{off}}$



VM导通: $U_i = L \frac{\Delta i_{on}}{T_{on}} \Rightarrow \frac{U_o}{U_i} = \frac{-p}{1-p}$

$p < 0.5$ $|U_o| < |U_i|$ 降压
 $p > 0.5$ $|U_o| > |U_i|$ 升压

$U_o = \frac{-p}{1-p} U_i$

$p \uparrow, U_o \uparrow$ $p \rightarrow 1, U_o \rightarrow \infty$

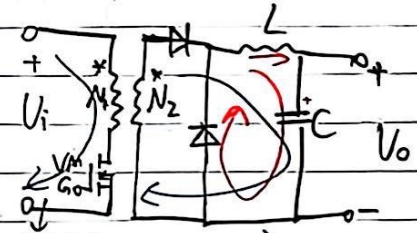
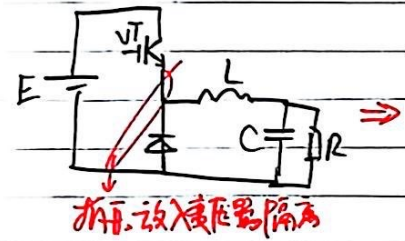
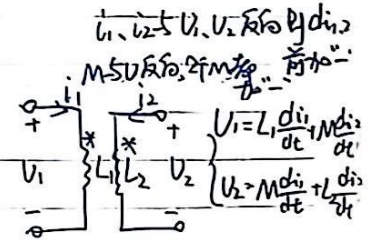
1°, 2°, 3° 可N匝并联 eg. Boost $\rightarrow$ 电压升

4° 单端正激

Buck可画为:

变压器常开:

$\frac{U_1}{U_2} = \frac{N_1}{N_2}$
 $\frac{i_1}{i_2} = \frac{N_2}{N_1}$
 $\frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2$



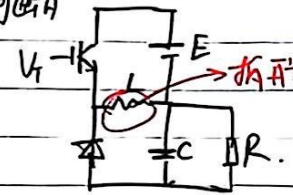
VM导通: $U_2 - U_o = L \frac{\Delta i_{on}}{T_{on}}$
 $\frac{U_i N_2}{N_1} - U_o = L \frac{\Delta i_{on}}{T_{on}}$

VM关断: $-U_o = L \frac{\Delta i_{off}}{T_{off}}$

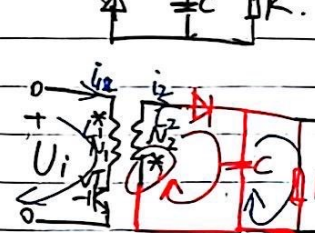
$\frac{1}{p} \chi = \frac{U_i N_2}{U_o N_1} \Rightarrow \chi = p \frac{U_i N_2}{U_o N_1}$

5° 单端反激:

* Buck-Boost可画为



$\frac{\Delta i_{on}}{\Delta i_{off}} = -\frac{N_2}{N_1}$



VM导通: $U_i = L \frac{\Delta i_{on}}{T_{on}} \Rightarrow \frac{U_i}{U_o} = \frac{1}{\frac{\Delta i_{on}}{\Delta i_{off}} \frac{T_{on}}{T_{off}}}$

VM关断: $-U_o = L \frac{\Delta i_{off}}{T_{off}} \Rightarrow \frac{U_o}{U_i} = \frac{+p}{1-p} \frac{N_2}{N_1}$

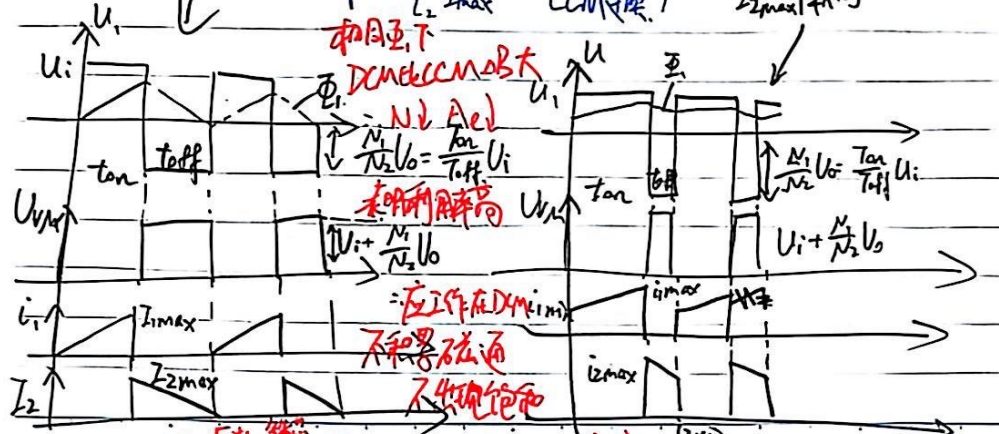
$i_1 = \frac{U_i}{L_1} t$ $I_{1max} = \frac{U_i}{L_1} t_{on}$

$i_2 = I_{1max} - \frac{U_o}{L_2} t$ $t_{off} \Rightarrow \frac{U_o}{L_2} I_{2max}$ DCM断使 $p \downarrow, I_2$ 降到0

$n = \frac{N_1}{N_2}$

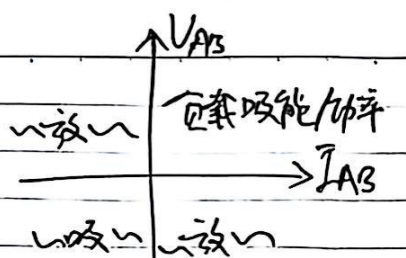
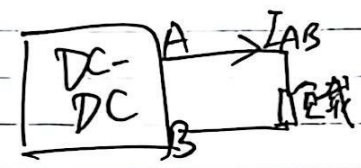
$t_{off} = \frac{U_o}{L_2} I_{2max}$ 临界

$t_{off} < \frac{U_o}{L_2} I_{2max}$ CCM连续 $p \uparrow, I_{2max}$ 降不到0



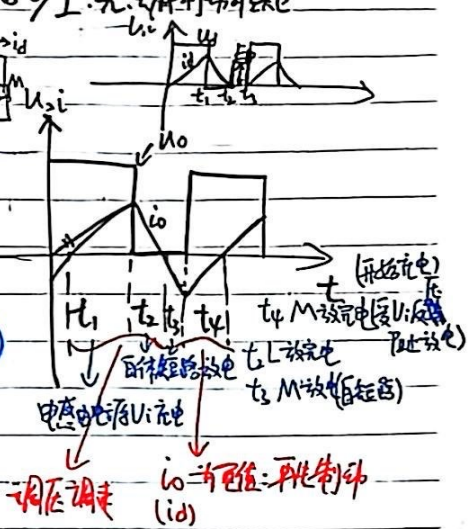
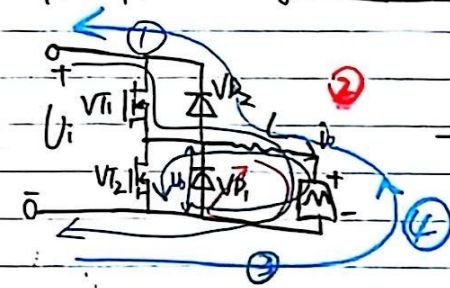
$U = N \frac{d\Phi}{dt} = N \frac{d}{dt} \left(\frac{1}{A} \int B dt \right) \Rightarrow \int u dt = N A \Delta B$

6° * DC-DC 四象限运行

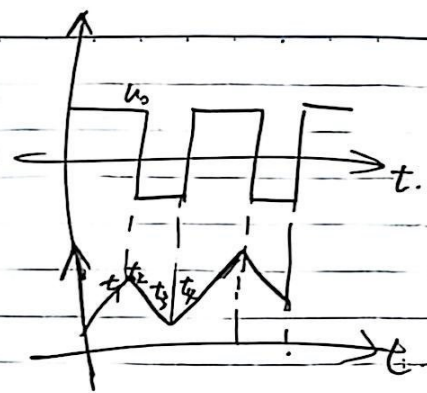
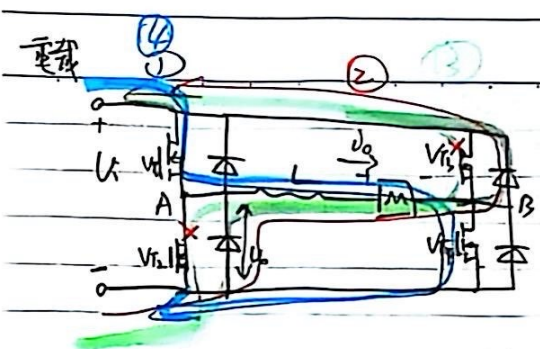
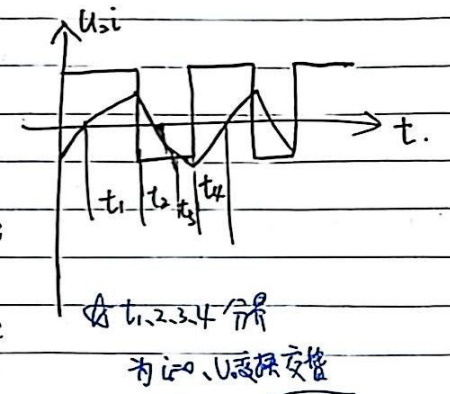
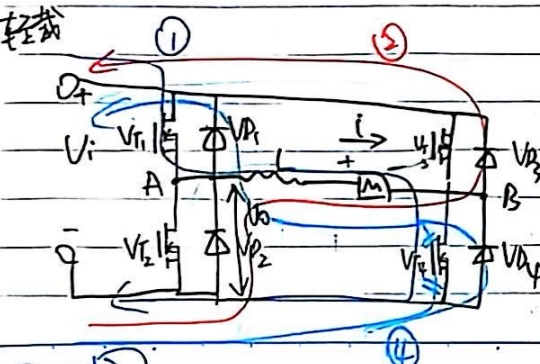


一般斩波如 Buck 只能调速. 无法反向 V_i . 无法有制动的转矩

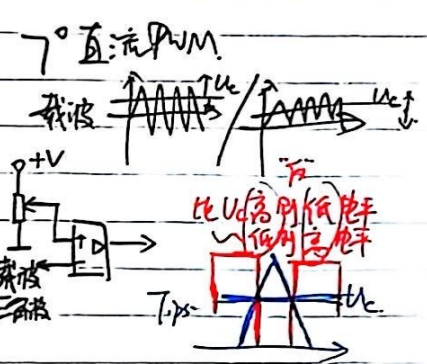
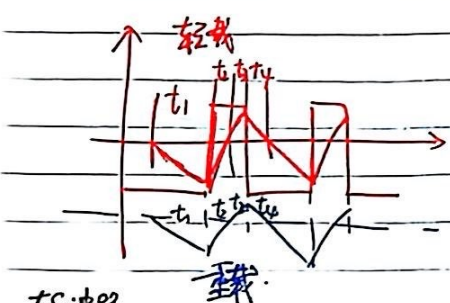
* 两象限斩波器



4象限斩波器



* 改变电压占空比可使电机反转



斩波器	非受限 (工频转变频)	受限 (工频转变频, $U_d \uparrow$ 外特性非线性)
☆ 分频概念		
不可逆输出 (电机一个方向)	两象限	单象限
可逆输出 (电机可2个方向)	四象限	

Chapter 7 无源变与变频

DC $\rightarrow$ AC 正弦?
 脉冲叠加 $\Rightarrow$ "多重化"
 不等幅 PWM
 等幅 PWM $\checkmark$
 (不等宽) $\downarrow$

S 等效 T_h: 冲量 $\Rightarrow$ 窄脉冲面积 $\downarrow$
 效果 $\Rightarrow$ 响应波形相同

生成 PWM 计算法 调制法 $\checkmark$

指定谐波消除法: 计算出能除谐波的脉冲相位 $\Rightarrow$ 开关通断

PWM 要全控器件!

SPWM: 载波: 三角波(恒幅) 改变占比 $\Rightarrow$ 改调制 sin 的幅值
 调制波: sin 正弦

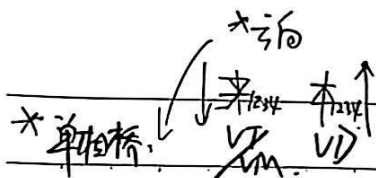
载波 $\uparrow$ 越精确

调制 sin 与 输出 sin 同频 (调制就是调制波的频率)

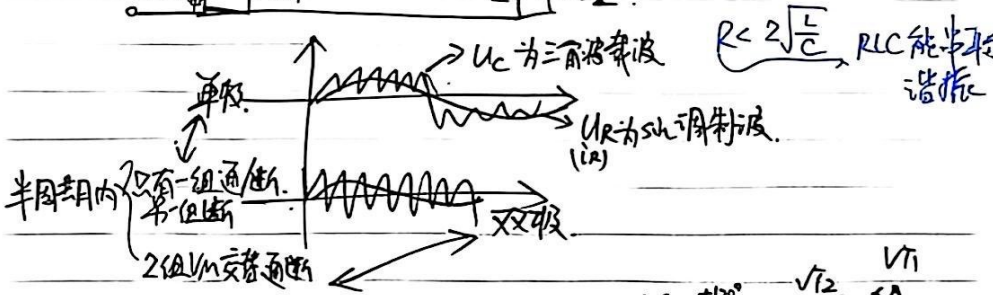
调制 sin 幅值 $\uparrow$, 输出 sin 幅值也 $\uparrow$

优点: 输出接近正弦谐波 $\downarrow$
 DC 由本侧整流, AC 电网的功率因数与逆变器输出 U, f 无关

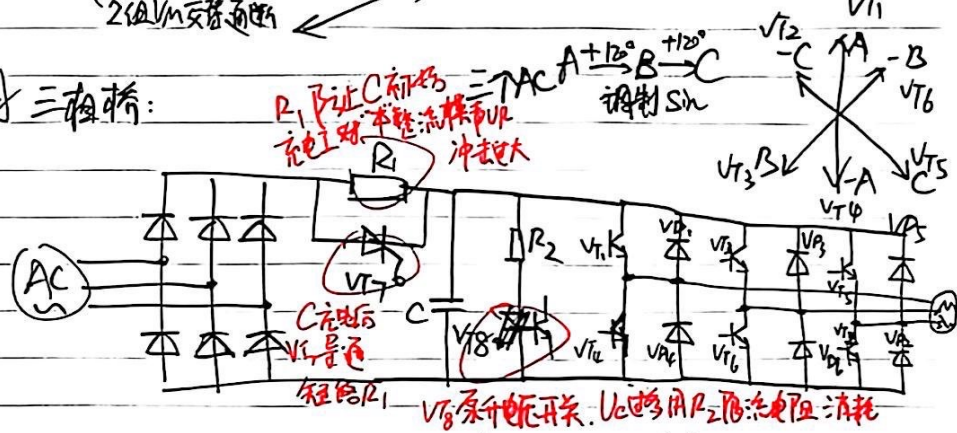
f, U 可分别/同时在逆变器内快速调节



C 为补偿电容
 使 RLC 串联谐振在一个周期内持续进行



三相桥:

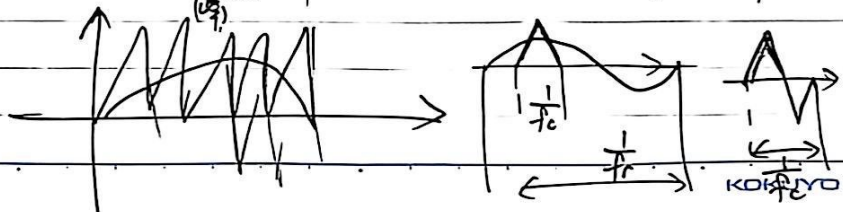


$U_{T1} \sim 6$ 间隔 60°
 $\frac{1}{2}, \frac{4}{2}$ 差 180°
 $\frac{1}{3}, \frac{5}{3}$ 差 120°
 $\frac{2}{3}, \frac{6}{3}$ 差 120°

需防止直通短路!

定义: 1. 调制度 $M = \frac{\sin \text{幅值}}{\Delta \text{幅值}} < 1$

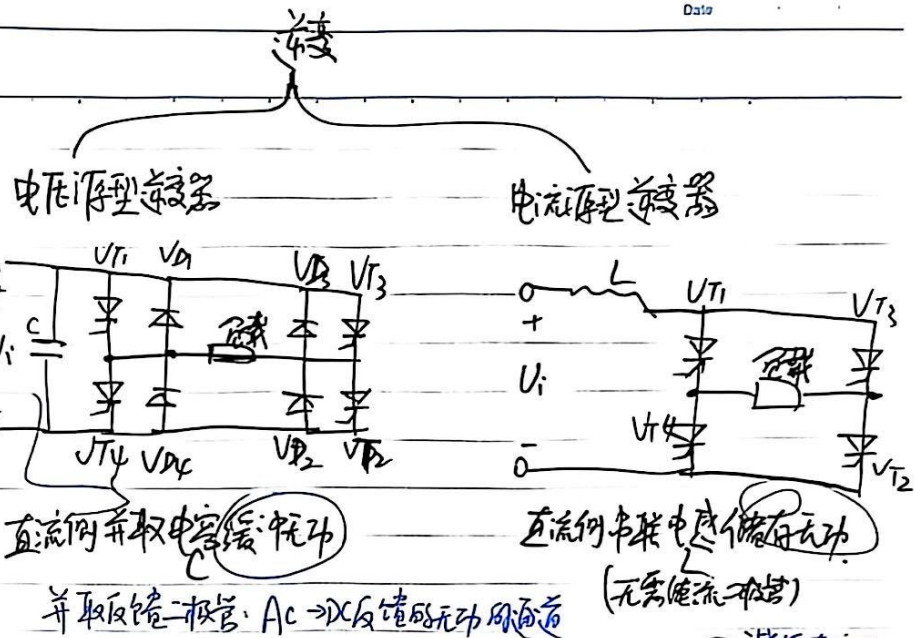
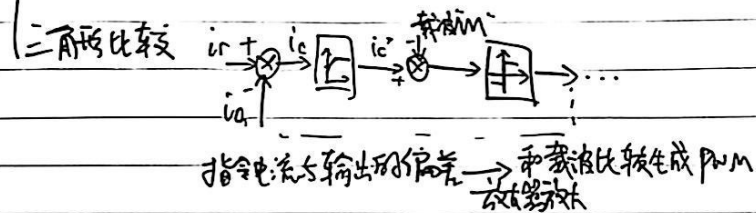
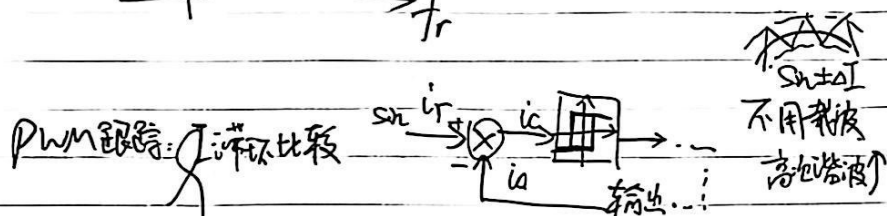
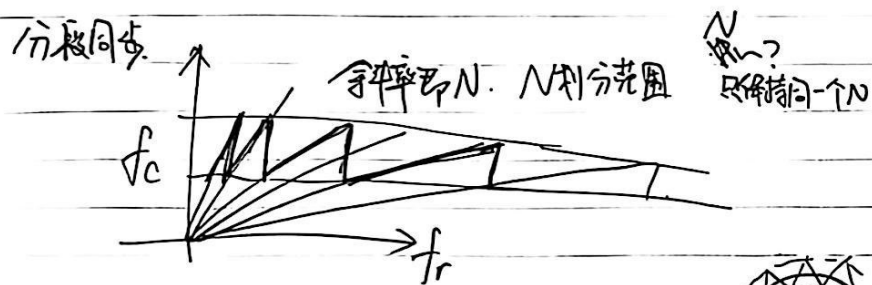
2. 载波比 $N = \frac{f_c \text{ 载波}}{f_r \text{ 调制}}$



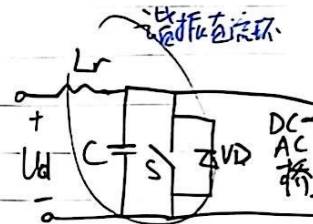
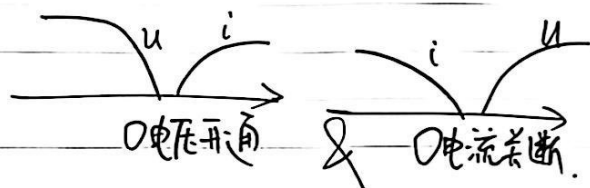
载波 $f_c \uparrow$, 谐波 $\downarrow$

但 f_c 不宜过高: 开关/损耗
对 U, i 大变化 $\Rightarrow$ 电磁干扰
对 C, L 冲击 I & 尖峰电压

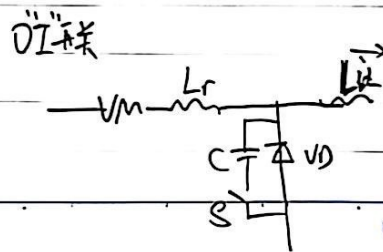
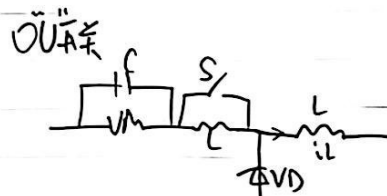
同步调制: 变 f 时, N 不变为常数
异步调制: N 不为常数



Chapter 8: 软开关 (谐振开关技术)



* 原理: 为使 U, i 为0, 利用 L, C 串联谐振电路实现, 形成正弦波。
在过0时接通或断开开关, 进行能量切换。



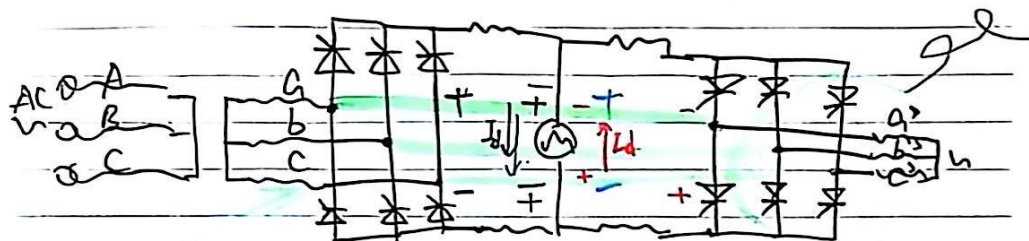
Chapter 9 应用

1. 电机调速 直流电机调速逻辑

AC → DC 调α. 改电枢V & I. → 改n, T.

外接AC时: 电枢反接可逆线路

反并联: 两组整流电源独立, 只可同时一组工作. 否则+与-短路



交叉连接

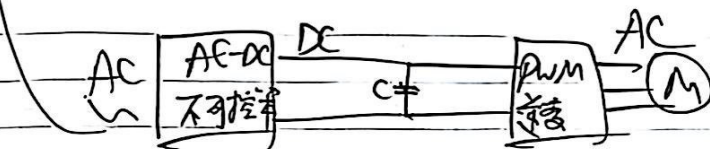
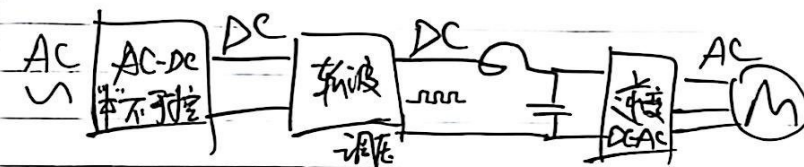
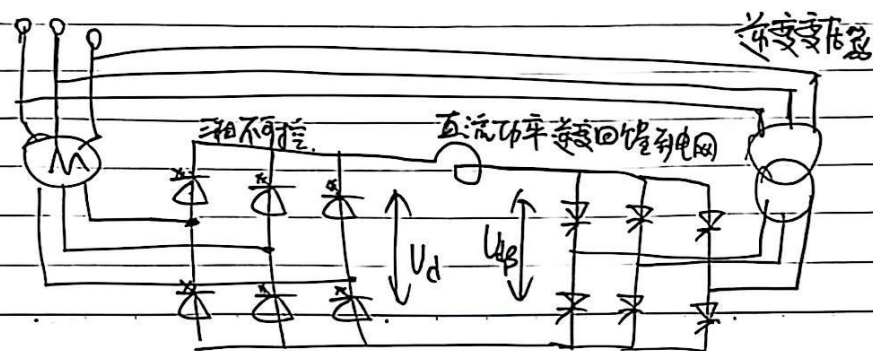
正组整流 正接拖车

正组续流 回馈制动

反组整流 反接拖车

反组续流 回馈制动

串级调速



2. 发电

